

Algebra II with Trigonometry

Chapter 5.6

Mental Math (GPT-5.4)

Saved tutoring response with MathJax rendering. Original PDF file:

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- 1. Recognizing the Rule** If two sides of a triangle are 5 and 7, and the included angle gets larger, does the third side get larger or smaller?
- 2. Right Triangle Check** A triangle has sides 6, 8, and 10. What is the largest angle?
- 3. Acute or Obtuse?** A triangle has sides 5, 6, and 8. Is the angle opposite the side of length 8 acute, right, or obtuse?
- 4. Isosceles with Included Angle** Two sides of a triangle are both 10, and the included angle is 60° . What is the third side?
- 5. Straight Comparison** Two triangles each have sides 8 and 8. One has included angle 30° , the other has included angle 120° . Which triangle has the longer third side?
- 6. Heron's Formula Setup** A triangle has side lengths 5, 5, and 6. What is its semiperimeter?
- 7. Area from Familiar Sides** What is the area of a triangle with side lengths 3, 4, and 5?
- 8. Bearing Language** What compass direction does the bearing $N 40^\circ E$ describe?
- 9. Travel Path Estimate** A car goes 30 miles east, then 40 miles north. About how far is it from its starting point?

10. **Kite Lines** Two kite strings are each 100 ft long. If the angle between them is 60° , how far apart are the kites?